
Programming with Python

BASIC CONCEPTS 2

Working with files

Python's "with open(...) as ..." Pattern

Getting a Directory Listing

Getting File Attributes

Making Directories

Deleting Files and Directories

Copying, Moving, and Renaming Files and Directories

Python's "with open(...) as ..." Pattern

❑ Read file

```
with open('data.txt', 'r') as f:  
    data = f.read()
```

❑ Write data to a file

```
with open('data.txt', 'w') as f:  
    data = 'some data to be written to the file'  
    f.write(data)
```

Getting a Directory Listing



Directory Listing

- **os.listdir()** - Returns a list of all files and folders in a directory
- **os.scandir()** - Returns an iterator of all the objects in a directory including file attribute information
- ...

Getting File Attributes



`os.stat(path)`

- `st_mode` – protection bits.
- `st_ino` – inode number.
- `st_dev` – device.
- `st_nlink` – number of hard links.
- `st_uid` – user id of owner.
- `st_gid` – group id of owner.
- `st_size` – size of file, in bytes.
- `st_atime` – time of most recent access.
- `st_mtime` – time of most recent content modification.
- `st_ctime` – time of most recent metadata change.

Making Directories

Creating a Single Directory

```
import os
os.mkdir('sample_data/example_directory/')
```

Creating Multiple Directories

```
import os
os.makedirs('sample_data/fpt/fa')
```

Deleting Files and Directories

Deleting Files in Python

```
import os
data_file = '/content/sample_data/mnist_train_small.csv'
os.remove(data_file)
```

Deleting Directories

```
import os
trash_dir = '/content/sample_data/test'
try:
    os.rmdir(trash_dir)
except OSError as e:
    print(f'Error: {trash_dir} : {e.strerror}')
```

Copying, Moving, and Renaming

Copying Files in Python

```
import shutil

src = '/content/sample_data/california_housing_train.csv'
dst = '/content/sample_data/fpt/fa'
shutil.copy(src, dst)
```

Copying Directories

```
import shutil
shutil.copytree('/content/sample_data/fpt/fa', 'data1_backup')
```


Copying, Moving, and Renaming

Moving Files and Directories

```
import shutil
shutil.move('/content/sample_data/fpt/fa/', '/content/')
```

Renaming Files and Directories

```
import os
os.rename('/content/data1_backup/california_housing_train.csv', '/content/
data1_backup/california_housing_train_copy.csv')
```

Thank you
